

4.5 Detailed specific example: Türkiye



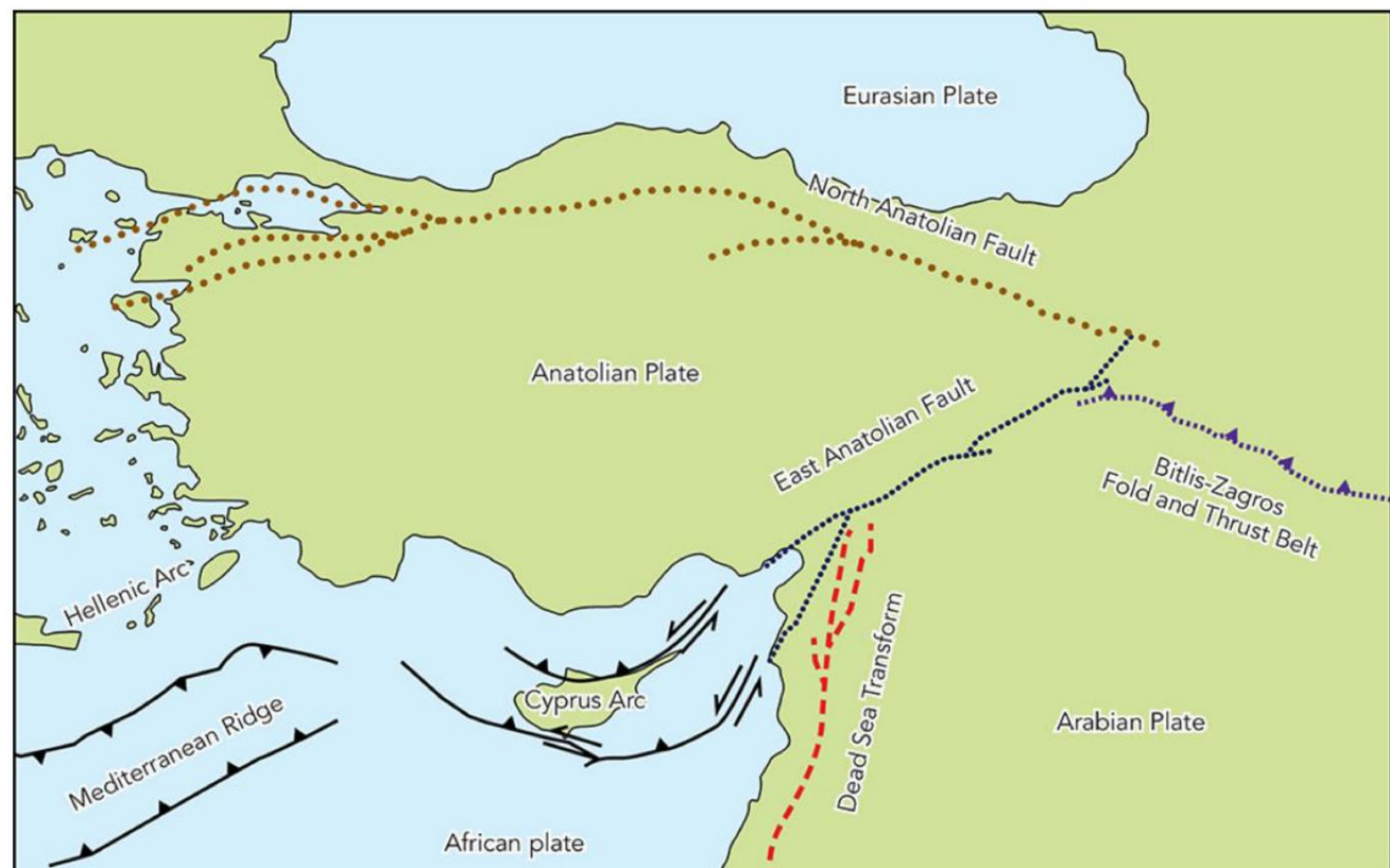
On 6 February 2023, at 4:17am local time, a magnitude 7.8 earthquake occurred in southern and central Türkiye and northern and western Syria. The epicentre was 37 km west-northwest of the town of Gaziantep, Türkiye, and the focus was at a depth of between 5 and 10 km. At the epicentre, the quake measured XII on the Mercalli scale. Nine hours later, a magnitude 7.7 earthquake followed, 95 km north-northeast of the first one. There were more than 30 000 **aftershocks** following these earthquakes. It was the deadliest natural disaster in modern history. This detailed specific example focuses on the impact on Türkiye.

Causes: Anatolian plate boundary movement

The earthquake occurred where complex plate boundary activity is taking place. Türkiye sits on the Anatolian plate. The Anatolian continental plate is a wedge trapped between three other plates moving in different directions.

With the Africa plate moving north and slowly subducting under the Eurasian plate, and the Arabian plate moving northwest, this is effectively squeezing the Anatolian plate that Türkiye sits on. The Anatolian plate is a continental plate that is separated from the Arabian plate in the south and Eurasian plate in the north, by the East Anatolian and North Anatolian fault lines.

The North Anatolian Fault line has historically had the strongest earthquakes. However, the 2023 earthquake occurred along the East Anatolian fault line. This released large amounts of energy as one plate moved past the other. Research suggests that the whole Anatolian plate is rotating counterclockwise. It is being pushed west by the Arabian plate but is blocked from moving north by the Eurasian plate (Source A).



Source **A**: A map of the plate boundaries on which Türkiye is located

- a Describe the location of the plates and plate boundaries around the plate that Türkiye is on.
- b Explain the cause of the earthquake on 6 February 2023.

Impacts

The earthquake caused widespread damage over an area of approximately 350 000 km². Figure 4.4 shows some of the main impacts caused by the earthquake.

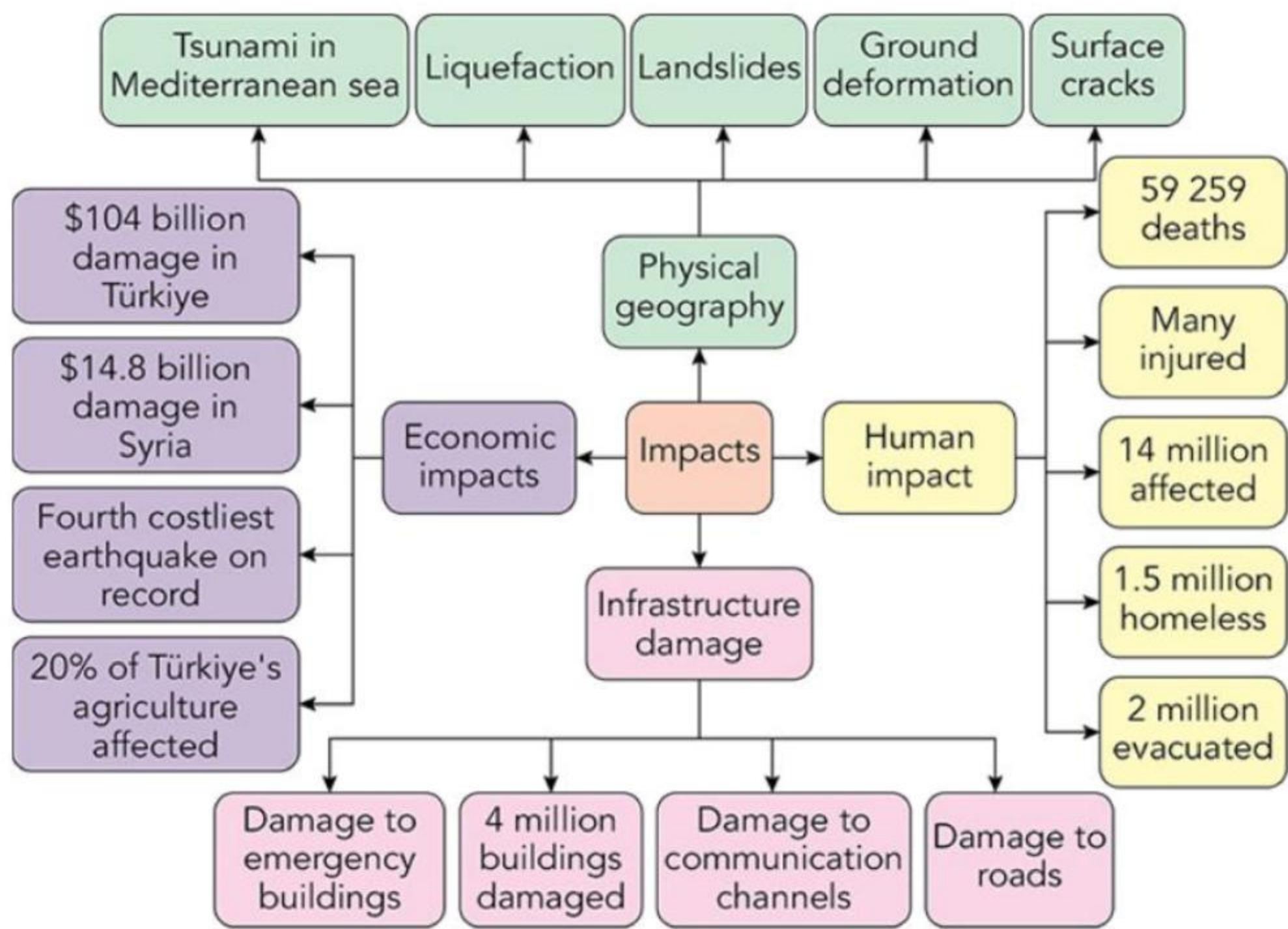


Figure 4.4: Impacts of the Türkiye-Syria earthquake

► Image Description



Source B: A building that suffered damage during the 2023 earthquake

TASK 2: Study Source B

- a** Liquefaction caused buildings to collapse. Explain what is meant by liquefaction and explain why it would cause buildings to collapse.
- b** In pairs, discuss what you observe in this building. Do you think it would be safe for emergency personnel to search the inside of this building? Explain your answer to each other and then discuss this with the class.

Responses

The response to the Türkiye-Syria earthquake was both local and international. The network of seismographs around the world means that knowledge of both the time and scale of the initial earthquake was known almost immediately, all around the globe. This meant that international teams were able to start preparing for a response immediately.

There are both primary and secondary responses to a natural disaster. The immediate, primary responses of this earthquake included:

- The government of Türkiye declared a state of emergency to make funds available to assist those in need of urgent assistance.
- Mosques, shopping malls, stadiums, schools, and community centres were used to shelter people unable to return home and at great risk due to the freezing winter storm.
- Mobile kitchens were distributed across affected areas to feed people.
- Free flights were offered to evacuate people.
- Local emergency personnel immediately started emergency action, rescuing approximately 8000 people from the rubble within the first week.
- Türkiye requested assistance from the international community, including emergency personnel, aid, clothing, medicines, food, and shelter.

One of the most significant challenges was the initial rescue and relief effort. This was difficult because of damage to transport routes and communication channels. The infrastructure damage made it difficult to contact and reach people in the worst affected areas. Additionally, as the earthquake happened during Türkiye's winter, the weather made environmental conditions more challenging and dangerous.

Long-term secondary responses included state commitments to:

- pay compensation for lost homes, and make funds available for relocation and rent assistance
- rebuild houses and other buildings in areas damaged by the earthquake
- rebuild roads and communication systems.

Source C: The number of emergency personnel that responded to the disaster

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Emergency category	Number of people
Search and rescue	60 000
Health workers	5000
NGO volunteers	30 000
International personnel	141 000

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TASK 3: Study Source C

- a **Calculate** the total number of emergency personnel that responded to the earthquake.
- b **Convert** the figures in Source C into percentages.
- c Use the percentages to **plot a pie chart**.
- d **Using data from the pie chart** and your calculations, describe the distribution of emergency personnel by category.

TIP

When manually drawing a pie chart, divide the chart using marks every 36° around the circle. Each mark represents 10% of the circle. This makes it easier to work out where your lines need to be placed to correctly plot the pie chart.

Managing the impacts of earthquakes in Türkiye

The Turkish government announced a plan to build 488 000 earthquake-resistant homes in the affected region within the first year. It also pledged to build several non-residential buildings such as schools and hospitals. The reconstruction plan included the strengthening of existing buildings that would sustain light, non-structural damage. In addition, it planned on redeveloping roads and bridges that were damaged. The total reconstruction costs are expected to exceed \$100 billion. Full recovery is expected to take many years.

Rebuilding after a disaster like an earthquake often prioritises speed to house affected people as quickly as possible. This can result in building standards not being met and low-quality housing being constructed. However, rebuilding done using sustainable methods can provide both short and long-term solutions for residents.

For reconstruction to be sustainable, it should include community input and integrated urban planning. An integrated strategy looks at the needs of local people and methods for improving the earthquake resistance of new and existing buildings, to make them safer over the long term. Many of the buildings that collapsed in Türkiye were built before 1999 (when new building codes which included earthquake resistance, were implemented).

During construction, officials need to carefully check that strict building codes are met. This prevents collapsed buildings being placed by low quality structures with the same risks. Reconstruction can include strengthening and retrofitting buildings that are not up to code, but did not collapse, to make them safer.

To be truly sustainable, the newly built housing should also take climate change into consideration. The southern and central region of Türkiye is expected to experience more heatwaves and droughts in the future. Therefore, the buildings also need to take ventilation and insulation into consideration. In addition, with such widespread damage, there is an opportunity to rebuild the areas to minimise congestion and pollution that results from the modern way of life in cities. This type of reconstruction requires the focus to shift from the individual building to town planning. It needs to consider green areas, public transport, and mixed residential and commercial areas to reduce commuting distances and lower pollution levels.

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Source D: A map of the Türkiye-Syria region showing the earthquakes and aftershocks that were experienced along the East Anatolian fault line

TASK 4: Study Source D

- a** Aftershocks are smaller earthquakes that occur after a large earthquake. Considering that this earthquake was caused by two plates slipping along a fault, describe the distribution of the aftershocks shown on Source D.
- b** This earthquake occurred very early in the morning. Do you think the death toll would have been different if it had happened during the day? Discuss in pairs and then explain your decision with reasons, to a larger group or the class.